

Future City of Bangladesh: IoT Based Autonomous Smart Sewerage and Hazard Condition Sharing System

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Abstract—With the influence of science, the way people addressed challenges in the past has now changed. It's easier to make "stuff" smart now-to-day with the advancement of IoT. Thus, with the advancement of smart infrastructure and the emergence of the Internet of Things (IoT), it is possible to calculate real-time flood prediction in Bangladesh's sewerage system and also to track the ecosystem of underground tunnels easily and efficiently. This study represents the development of a sewerage system for an existing sewerage system in Bangladesh that can predict sewerage system overflow and implement a hazard condition monitoring system prototype using IoT technology by calculating risk factors such as calculating the amount of toxic gases in underground tunnels. We should save the lives of the employees who work in caves, contact workplaces and manhole underground work.

Keywords—component IoT, Sewerage System, FSR, valve, Mqtt broker, topic, MQ135, NodeMCU, Arduino.

I. INTRODUCTION (HEADING 1)

One of the most bustling vocabulary of this age is the Internet of Things (IoT) (Jaykishan Panchal) [1]. IoT will attract interest and reach out to people as it allows sensor nodes with multiple control capacities to provide additional conventional contact facilities [2]. By Arkady Zaslavsky et.al [3], by Saifuzzaman et.al. [4,5,6] and by Siddique [7] The Internet of Things enables the Internet to reach into the natural world and link physical objects to the Internet. To make things simpler, this project integrates IoT with the modern world. Gartner [8] notes that 25 trillion items will be connected to the internet and will be used by 2020. This suggests that more than twice the world's population would have wired devices. There are several ways to implement IoT in real life as monitoring devices and automation technology. By linking distant and remote areas, these networks can effectively provide better support and problem solving. Two modes of problem definition are the subject of the IoT application. It is important to solve the first specification style problem. It may be described as a pain reliever. The other is where a solution acts as a utility with value added. It can be graded as a vitamin. The IoT-based Smart Sewerage Grid focuses primarily on Wikipedia [9] from Bangladesh's Dhaka

City. This project aims to tackle the latest vitamin and pain killer epidemic. According to Population 2017[10], about 18 million people live in Dhaka. Throughout the monsoon season, almost 80% of the total annual precipitation of 1,854 millimeters occurs [11]. The city suffers terrible waterlogs in the rainy season due to having a very weak drainage system. The source behind this bad drainage system is clogged drains and channels. Waterlog as seen in Fig.1 in the rainy season.



Fig. 1. Dhanmondi 27 road after a heavy rainfall in Dhaka (Daily Star, 2015)

Source: Unusual rain: Climate change to blame – The Daily Star (September, 2015).

In addition, there is no predetermined blockage detection. When it happens and somebody tells the particular authority, the best way to minimize the waterlog is. According to IPS, 50-60 percent of Dhaka city could go underwater within 10 years without taking proper action [12]. Smart Sewerage System IoT bases are planned to be applied to improve this in the current sewerage system. It can ensure the upcoming blockage is predetermined and tends to avoid waterlog long before it occurs. The up-and-underground sewage system connection is seen in Fig. 2.

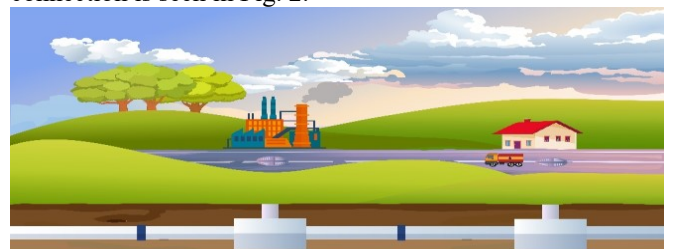


Fig. 2. Up ground and underground connection

The size of a manhole depends on the population density of a valued area, according to Dhaka Wasa [11], and each manhole has a fixed size. Each manhole contains a normal amount of water, based on the scale of the manhole and population density. The water level of the manhole is rising due to heavy rain. If the water level rises so high, the imminent blockage is predetermined. A fixed unit would be mounted under the cover of a manhole in the IoT Based Smart Sewerage Scheme. This software is used to scan the manhole's water height, equate it to the usual water height, and transmit the details to the dashboard. The smart sewerage system dashboard is shown in Fig. 3.

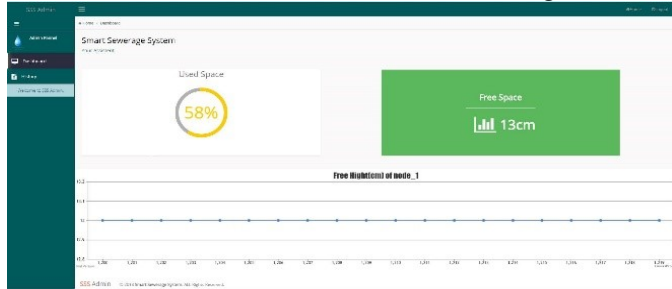


Fig. 3. Dashboard of Smart Sewerage System

Database receives sensor data and converts into free space and used space using following formula:

$$\text{Blank Space} = \frac{(\text{Full Length} - \text{Water Height})}{\text{Full Length}} 100\%$$

The manual cleaning of clogged sinks and manholes has another downside. The drains and manholes are cleaned only once a year, according to Dhaka Wasa [13]. It continues to create poisonous gases within the closed manholes as the manhole stays sealed for a long period of time. A lot of deadly injuries occur as the sweeper heads straight into the manhole without testing the inside environment. Ammonia (NH₃) is primarily responsible for the deadly climate. But only by testing the inner atmosphere of the covered manhole can this death be minimized and avoided. The sweeper will leave the manhole open for a while and go inside after a period when the air is dangerous or close to dangerous. It could take a few more hours to scrub, but it's going to save a human life. And he can safely go indoors without losing his own life if the air is secure. A lot of time is more important than time to forget about life. The Real Life Cleaning Manual is seen in Fig. 4.



Fig. 4. Manual cleaning of a manhole (Daily Mail, 2017)

Source: Spare thought for Bangladesh sewer cleaner who dives into filth to claw out blockages – Daily Mail (May 2017)

The handheld system known as a toxic level detector, which senses the environment of a manhole and warns the user whether the manhole is safe to clean or not, is another remarkable feature of this project. In order to detect the air condition, there are three LEDs and one MQ135. The healthy, alert and deadly condition is alluded to by the Orange, Yellow and Red LEDs. This initiative aims to pre-determine the

potential waterlog in a project overview and to ensure the welfare of sweepers. And before it takes shape, this initiative strives to reduce the crisis.

II. LITERATURE REVIEW

Hlodversdottir *et.al.* [14] showed their analysis was to consider the actual and future imminent possibility of flooding in the combined sewer system in the middle of Reykjavik. Nie *et.al.* [15] showed in their research work about the current case, projected and artificial climate scenarios containing (1) surface flooding (2) basement flooding and (3) combined sewer overflow is specified to reflect the adverse effects of climate change with regard to the potential outcomes of the Sewerage System. Weyrauch *et.al.* [16] and Passerat *et.al.* [17] presented the Sewerage System overflow problem scenario and made possible wastewater treatment plant method which was applied to the urban River Spree in Berlin, Germany. Hata *et.al.* [18] aimed analysis of the impact of rainfall events affecting viral load in surface waters infected by the Sewerage System and of the efficacy of molecular enteric virus detection methods. Grum *et.al.* [19] Starting with rainfall / runoff methods, recent advances in urban storm drainage are analysed, followed by discussions on combined sewage, drainage effects on water absorption, impact control, hydro informatics, regulatory systems, and conclusions. Chen *et.al.* [20] discussed about the overwhelmed Sewerage Grid, the surface flooding and showed a modern method to replicate & reduce the surface flow rather than the conventional model. In our proposed model, we plan to build a model that can be calculated by real-time estimation of flooding in the Bangladesh sewerage system. Abraham *et.al.* [21] focused on using IoT technology to construct the air & soil quality control system and from the real time details shown on a graphical user interface. From this idea, we are launching a prototype using IoT technologies and designing a hazard status management system. We will save the lives of staff who do underground work in caves, contact work sites and manholes by monitoring risk factors such as monitoring the amount of toxic gases in underground tunnels.

III. EXPERIMENTAL METHOD

IoT based Smart Sewerage System focuses on a major existing problem of Dhaka city. This problem is not visible until there is no rain.

A. Developed Work

Two big current challenges to be addressed are the subject of the whole project. The first is to detect the quantities of water and the second is to identify the dangerous situation. This separates the work produced into two sections of the work created.

1) *Fixed Device:* To predetermine the upcoming blockage, an ultrasonic sensor, FSR (Force Sensitive Resistor) and a customized valve are included in the fixed unit. The Servo Motor was used to personalize the valve. Water pressure is detected by the FSR and a signal is sent to the valve. The valve is used to protect the ultrasonic sensor from water and dust by covering and maintaining it. The pressure from the water is received when the water level rises to the cover of the manhole and reaches the FSR and the valve remains locked. A blocking signal at this stage is received by the MQTT Broker. While there is no strain, the valve stays open and the ultrasonic sensor

scans the height of the free manhole and sends it to the MQTT Broker. Both the fixed computer and the broker subscribe to the same subject for receiving results. After the MQTT broker receives the data, it sends it to the database and finally the results appear in the dashboard. For this approach, NodeMCU was used. In real life solution it will absorb power from lamppost and GSM will be installed in to detect location.

2) *Portable Device*: The handheld unit (toxic level detector) is used to safeguard the integrity of the sweeper. The MQ135 unit contains three LEDs.

MQ135 tests the consistency of the air at PPM. The unit can switch on green LEDs if the PPM-mg / l is 0.000-0.050, saying that the air is healthy. If the PPM-mg / l is 0.050-0.500, by saying the atmosphere is alert, the system would turn on yellow LEDs. And if the PPM-mg / l is 0.500 +, by stating that the air is lethal, the system will turn on red LEDs. For this solution, Arduino was used. Fig 5 displays the portable computer.



Fig. 5. Toxic Level Detector (Portable Device)

IV. TECHNICAL ARCHITECTURE AND SAMPLING

The technical architecture of the project is shown in fig. 6.

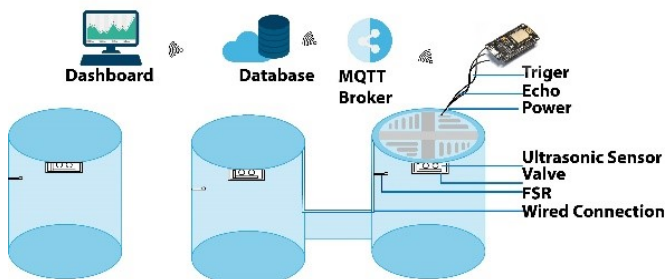


Fig. 6. Technical Architecture of Smart Sewerage System

This device helps users only by testing the dashboard to detect the imminent blocking. Typically, manholes have a set height and a standard height for water. If the height of the water abruptly rises so high, it reveals the upcoming blockage. The whole region has been broken into nodes for the sampling of this initiative. A particular position is indicated by every node. Red circles are classified as nodes in Fig 6. If the water level unexpectedly raises in one node, it is obvious that blockage happens in the prior node of the same node. The precise blockage location is observable as the nodes are separated according to the location. A consumer has to register with the system by supplying the necessary information to access the dashboard. Users can quickly view the dashboard after enrolment and have correct knowledge of

the current situation. In fig 7, the node sample is seen.



Fig.7. Node Samples

V. COST STRUCTURE

In the performance of the project, the cost structure plays an important part. We wanted to keep the prices as competitive as we could. The prototype cost estimation and real life are somewhat different. For LEDs, wires and appliances, there are a few extra expense fractions like those not included in the structure. Table I and Table II provide the cost structure for prototype and actual life.

Table I (COST STRUCTURE FOR PROTOTYPE)

Fixed Device			
SL	Equipment	Price (\$)	Price (BDT)
1	Ultrasonic Sensor	1.2	100
2	Servo Motor	1.5	125
3	Force Sensitive Resistor	7	580
4	NodeMCU	3.5	290
	Total	13.2	1095
Portable Device			
1	MQ135	1.69	140
2	Arduino	4.6	380
	Total	6.29	520

Table II (COST STRUCTURE FOR REAL LIFE SOLUTION) [*Customized]

Fixed Device			
SL	Equipment	Price (\$)	Price (BDT)
1	Ultrasonic Sensor	1.2	100
2	GSM	25.33	2100
3	Force Sensitive Resistor	0.12*	10
4	Solenoid Push Pull	0.5*	40
	Total	27.15	2250
Portable Device			
1	MQ135	1.69	140
2	Arduino	4.6	380
	Total	6.29	520

VI. IMPLEMENTATION AND RESULT

The IoT Based Smart Sewerage System prototype has already been built. A manhole proxy was used as a bucket. One bucket was completely covered with garbage for ammonia gas formation. If the PPM is too high for ammonia, the red LED triggers as an indicator of a deadly climate. In figures 8 and 9, the PPM ratio and the LED are shown.

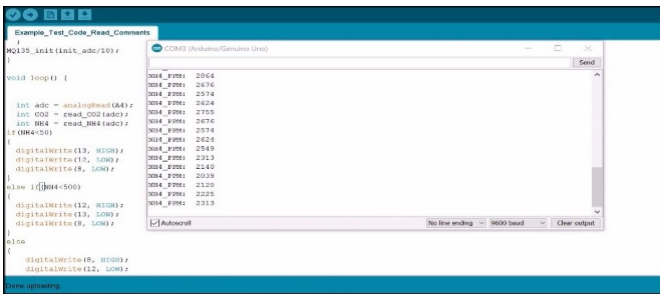


Fig. 8. PPM level of Ammonia (NH3)

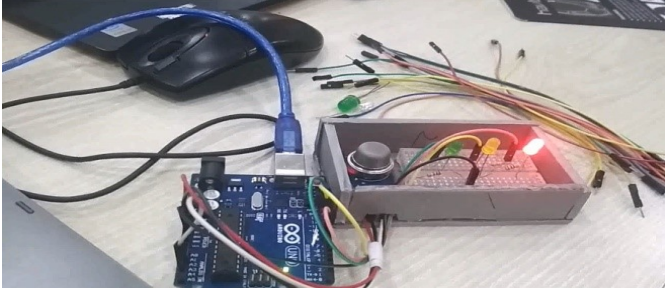


Fig. 9. Red LED turns on as the indication of deadly environment. The PPM level decreases after several cycles and the yellow and green LED decreases in a valued manner. Figures 10 and 11 demonstrate the safest PPM and LED.

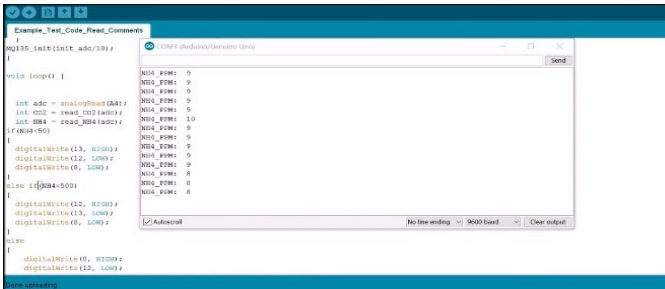


Fig. 10. PPM level of Ammonia (Safe)

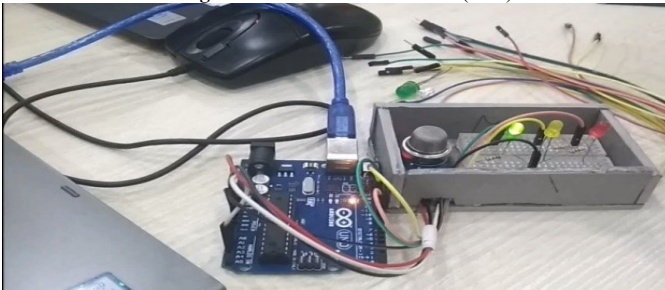


Fig. 11. Green LED turns on as the indication of safe environment. Another bucket was filled with water to detect water height and blockage. When water is minimum, the ultrasonic sensor senses the water level and displays the space and space used in the dashboard. The blockage signal to the dashboard is transmitted as the water reaches the threshold. Figure 12-15 displays the water volume and the dashboard results.



Fig. 12: Bucket Filled With Minimum Water



Fig. 13. Used Space and Free Space of the Bucket (Dummy Manhole)

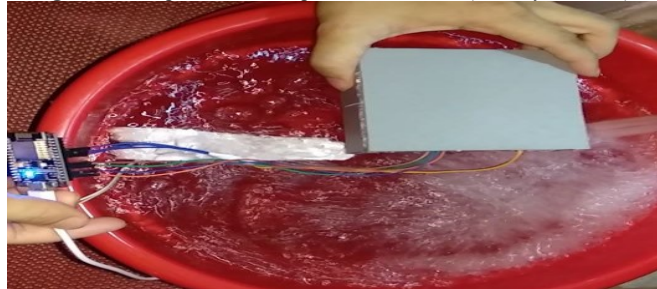


Fig. 14. Water Level Crossing the Threshold

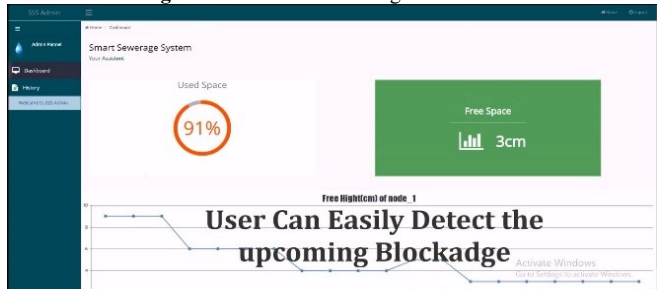


Fig. 15. Free Space and Used Space after Crossing the Threshold. Users should review sensor data along with historical node numbers. Figure 16 displays the background of sensor data.

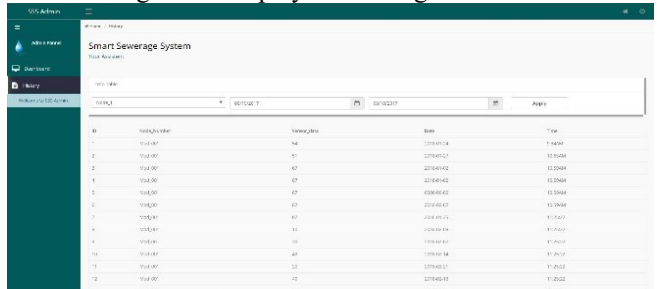


Fig. 16. Sensor data with Node Number

The final view of the prototype is shown in figure 17.



Fig. 17. Final Prototype of IoT Based Smart Sewerage System

VII. PERCEIVED CHALLENGES

In order to have a hassle-free life IoT connects more and more industries every day. In Bangladesh, it is not very easy to connect the sewer system with IoT. The project was completed with many challenges. The most challenging facts were existing facilities, the protection of ultrasonic sensors against dirt and water, data from a fixed system, the formation of ammonia in laboratory.

VIII. CONCLUSION & FUTURE WORK

In spite of all the difficulties, the IoT based Smart Sewerage and Hazard Condition Sharing solution can be used for the effective tracking of potential obstruction and the protection of sewer. However, once we connect lakes and canals to this project, the whole device design will not be successful. This research aims for a more holistic approach that takes the next work into account.

- Automated Blockage Removal
- Connect lakes and Canals
- Awareness Program
- Reward Program

Our plan for linking urban lakes and canals to the whole system is being developed. Dumping waste into water must stop. Education and awareness-raising campaigns will simultaneously aim for the right approach. The cleaning device is also manually reliant on the operation. The technique has already been developed for automatic blocking elimination. However, there needs to be more research. This is a complete, but step-by-step project.

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